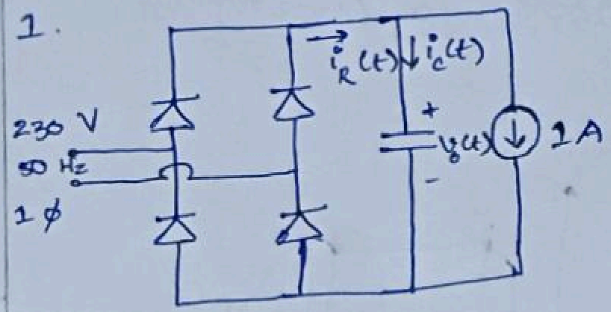
EE3203 : POWER ELECTRONICS

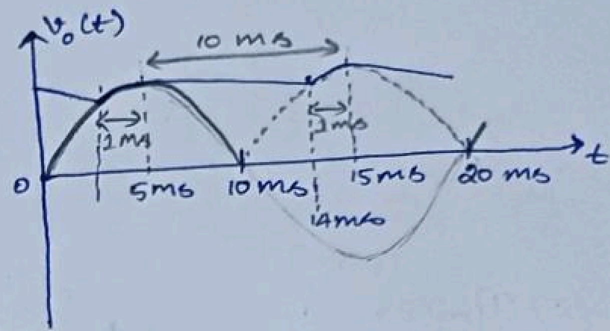
TUTORIAL 2 : POWER ELECTRONIC DEVICES AND FIGURE OF MERIT

1.



230 V (1-ϕ) ac i/p

$t_{diode, cond} = 1$ ms



(a). $i_c = C \dfrac{dv_c}{dt}$

$v_c = V_m - \dfrac{1}{C} t_c$

~~At t~~

for $t_{diode, cond} = 1$ ms,

$t_c = 10 \text{ ms} - 1 \text{ ms} = 9$ ms

$\Rightarrow v_c = V_m - \dfrac{1}{C}(9 \text{ ms}) \longrightarrow$ ① $\Big]_{@\ t = 14 \text{ ms},\ t_c = 9 \text{ ms}}$

At $t = 14$ ms,

$v_o = -v_{in} = -V_m \sin(\omega \times 14 \text{ ms}) \longrightarrow$ ②

① & ②, $V_m - \dfrac{1}{C}(9 \text{ ms}) = -V_m \sin(2\pi(50) \times 14 \text{ ms})$

$\Rightarrow C = \dfrac{9 \times 10^{-3}}{230\sqrt{2}\,(1 + \sin(1.4\pi))} = 565.3336\ \mu F$

① $\Rightarrow \Delta v_c = 230\sqrt{2} - \dfrac{9 \text{ ms}}{(565.3336\ \mu F)} = 15.9198$ V

Output peak-to-peak ripple, $V_{p\text{-}p} = 15.9198$ V

(b) $i_c(t) = C \frac{d}{dt} v_c(t)$

$= C \frac{d}{dt} (V_m \sin \omega t)$

$= \omega C V_m \cos \omega t$

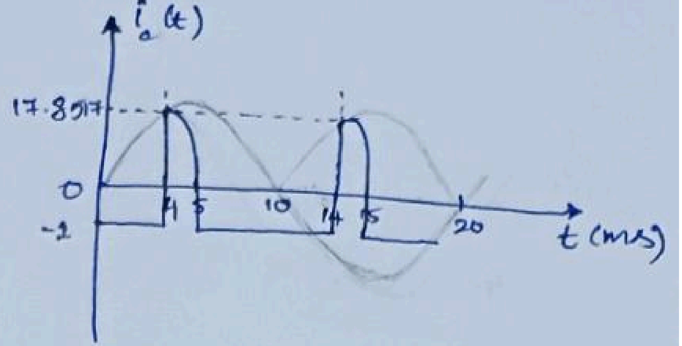
$= 2\pi \times 50 \times 565.3336 \times 10^{-6} \times 230\sqrt{2} \cos(100\pi t)$

$= 57.7694 \cos(100\pi t)$

~~At t = 5 ms,~~

$\cancel{i_{c,pk} = 57.7694 \cos(1.5708)}$

$=$



At $t = 4$ ms,

$i_c = 57.7694 \cos(1.2566)$

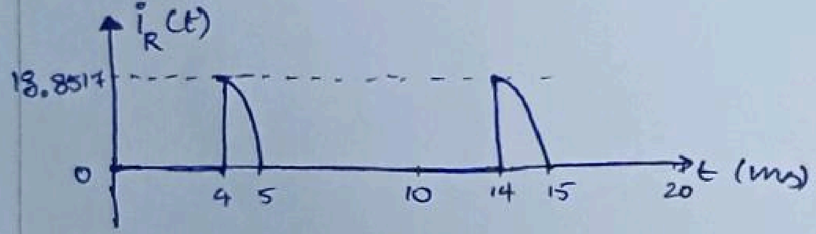
$= 17.8517$ A

At $t = 5$ ms,

$i_c = 57.7694 \cos(1.5708) \simeq 0$ A

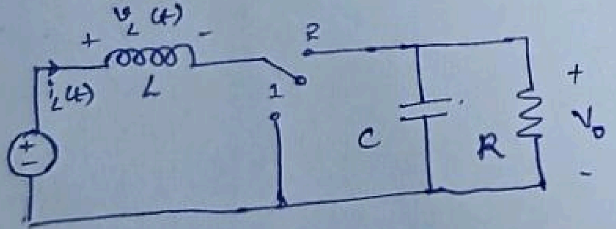
For $5\text{ms} \leq t \leq 14\text{ms}$, $i_c = -1$ A

(c). $i_R(t) = i_c(t) + 1$



(d). $C = 565.3336\ \mu F$ [Refer 1 (a)]

2.



$V_i = 15$ V

$L = 150\ \mu H$

$C = 470\ \mu F$

$R = 10\ \Omega$

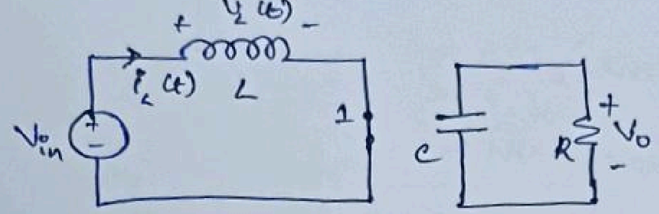
$V_o = 30$ V

$d = 0.5$

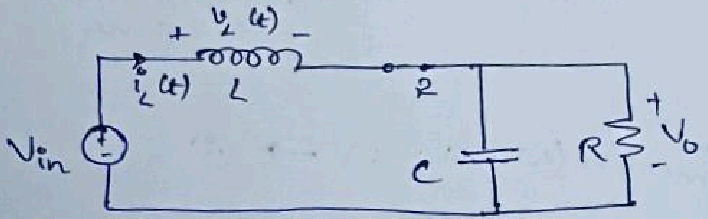
$f_{sw} = 20$ kHz

BOOST CONVERTER

(a) Position 1:



Position 2:



(b). $0 \leq t \leq DT_s$: $v_L = V_{in} = 15\text{ V}$

$DT_s \leq t \leq T_s$: $v_L = V_{in} - V_o = -15\text{ V}$

$V_o = \dfrac{V_{in}}{1-D}$; $P_{out} = P_{in}$

$V_o I_o = V_{in} I_{in}$

$\Rightarrow V_{in} I_{L,avg} = \dfrac{V_{in}}{1-D} \cdot I_o \Rightarrow \boxed{I_{L,avg} = \dfrac{I_o}{1-D}}$

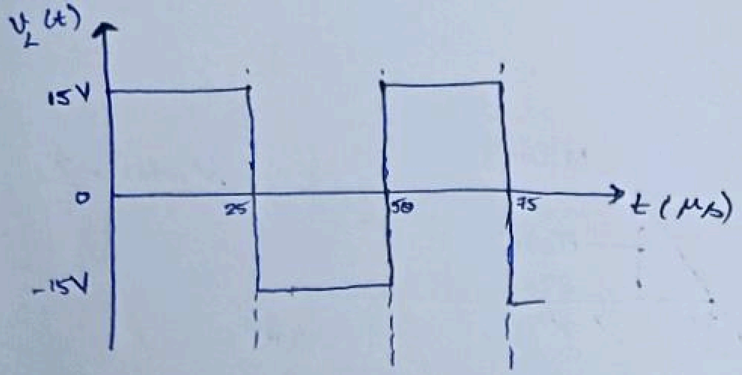
$I_o = \dfrac{V_o}{R} = \dfrac{30}{10} = 3\text{ A} \Rightarrow I_{L,avg} = \dfrac{3}{1-0.5} = 6\text{ A}$

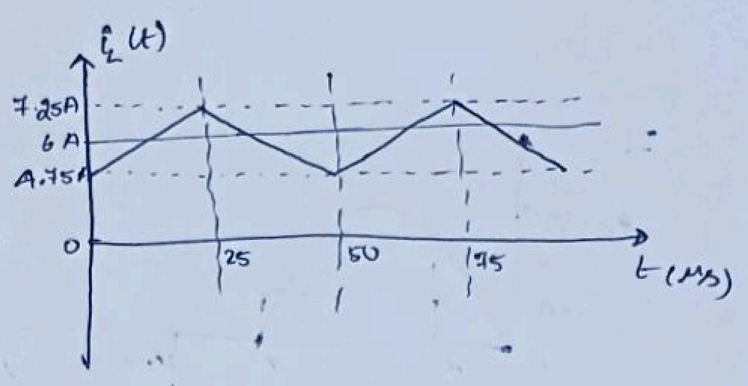
For $0 \le t \le DT_s$: $v_L = V_{in} = L \dfrac{\Delta i_L}{DT_s}$

$$\Rightarrow \Delta i_L = \frac{V_{in} D}{f_{sw} L} = \frac{15 \times 0.5}{20 \times 10^3 \times 150 \times 10^{-6}} = 2.5 \text{ A}$$

$$I_{L,max} = I_{L,avg} + \frac{\Delta i_L}{2} = 6 + \frac{2.5}{2} = 7.25 \text{ A}$$

$$I_{L,min} = I_{L,avg} - \frac{\Delta i_L}{2} = 6 - \frac{2.5}{2} = 4.75 \text{ A}$$

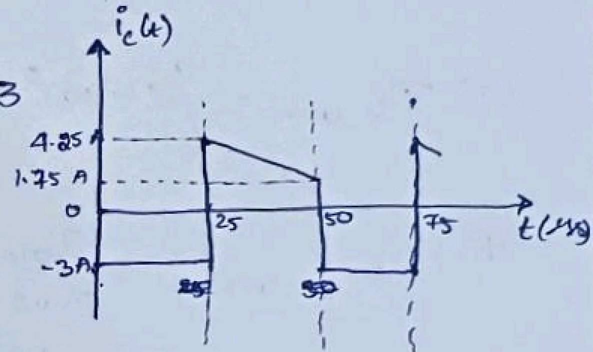




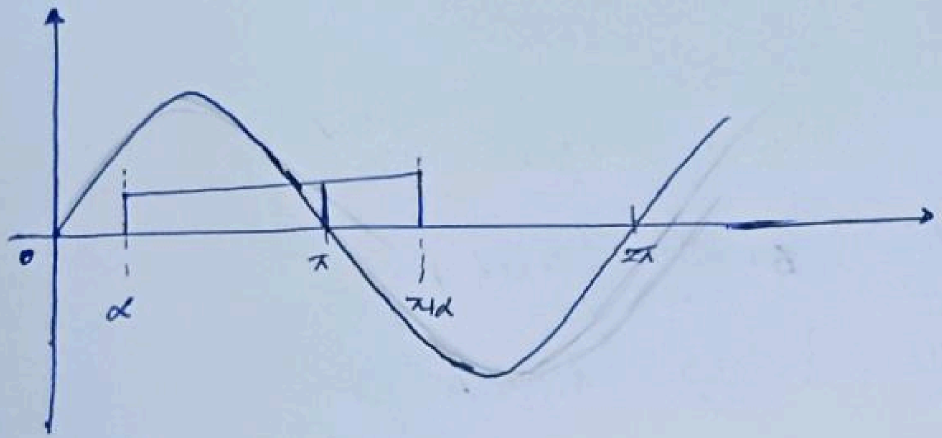
(c). $0 \le t \le DT_s$: $i_c(t) = -I_0 = -3\text{A}$

$DT_s \le t \le T_s$: $i_c(t) = i_L(t) - I_0 = i_L(t) - 3$

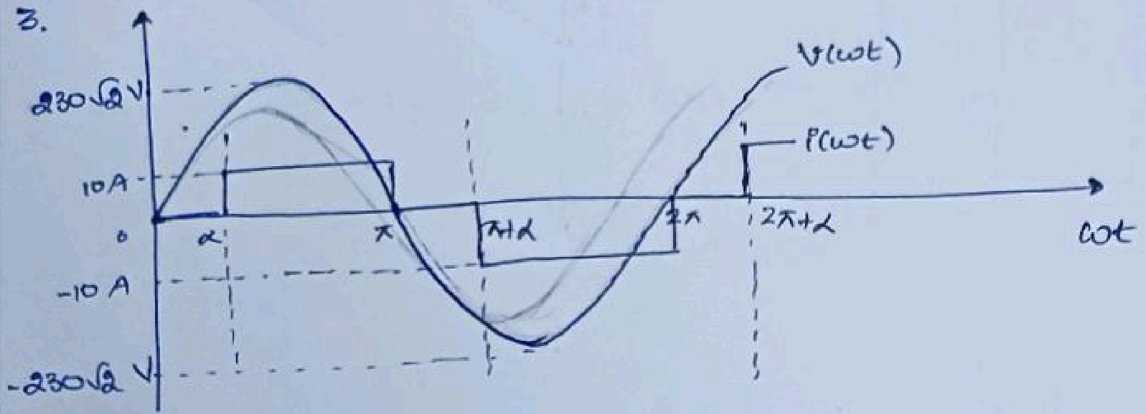
~~for $DT_s \le t \le DT_s$: $i_c =$~~



3.



3.



$\alpha = 60° = \frac{\pi}{3}$

(a). $I_{in,rms} = \left\{2 \times (10)^2 \times \frac{\pi - \pi/3}{2\pi}\right\}^{1/2} = 10\frac{\sqrt{2}}{\sqrt{3}} = 8.165$ A

(b). $I_{in,1(rms)} = \frac{4 I_{in,pk}}{\pi\sqrt{2}} \cos\frac{\alpha}{2}$

$= \frac{4(10)}{\pi\sqrt{2}} \cos 30° = 7.797$ A

(c). Crest Factor $= \frac{I_{in,pk}}{I_{in,rms}} = \frac{10}{8.165} = 1.2247$

(d). Displacement Factor $= \cos\left(\frac{\alpha}{2}\right) = 0.866$

Power Factor $= \frac{I_{in,1(rms)}}{I_{in,rms}} \cdot \cos\left(\frac{\alpha}{2}\right) = \frac{7.797}{8.165} \times 0.866 = 0.827$

(e). $P_{avg} = \frac{1}{2\pi}\int_0^{2\pi} v(\omega t)\, i(\omega t)\; d(\omega t)$

$$= \frac{1}{2\pi}\left[\int_{\alpha}^{\pi} V_m I_m \sin(\omega t)\, d(\omega t) + \int_{\pi+\alpha}^{2\pi} -V_m I_m \sin(\omega t)\, d(\omega t)\right]$$

$$= \frac{V_m I_m}{2\pi}\left[-\cos\pi + \cos\alpha + \cos 2\pi - \cos(\pi+\alpha)\right]$$

$$= \frac{V_m I_m}{\pi}(1+\cos\alpha)$$

$$= \frac{(230\sqrt{2})(10)(1+\frac{1}{2})}{\pi}$$

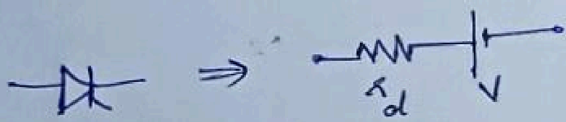
$\Rightarrow P_{avg} = 1.553$ kW

(f). $P_1 = V_{1,rms} I_{1,rms} \cos\phi_1$

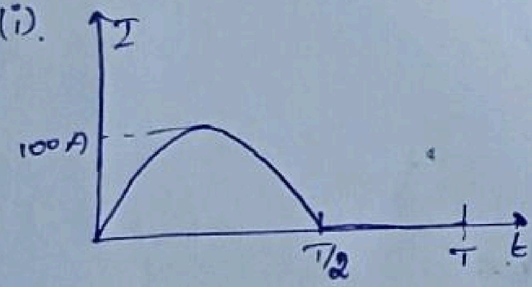
$$= \frac{230\sqrt{2}}{\sqrt{2}} \times \frac{4\times 10}{\pi\sqrt{2}} \times \cos 30^\circ . \cos 30^\circ$$

$\Rightarrow P_1 = \approx 1.553$ kW

4. <u>SCR:</u>



(i).



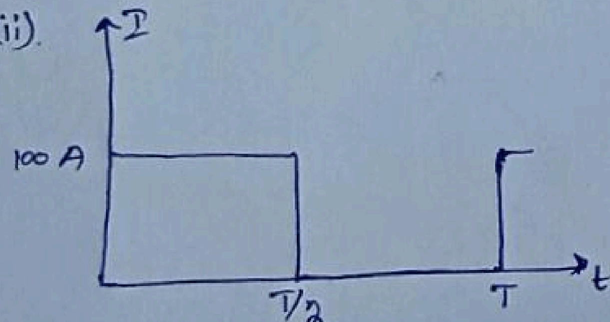
$P_{(?)} = 56.8$ W

$I_{avg} = \frac{I_m}{\pi}$

$I_{rms} = \frac{I_m}{2}$

$I_m = 100$ A

(ii).



$P = 100$ W

$I_{avg} = \frac{I_m}{2}$

$I_{rms} = \frac{I_m}{\sqrt{2}}$

$I_m = 100$ A

(a) (i). $I_{avg} = \dfrac{100}{\pi} = 31.831$ A

$I_{rms} = \dfrac{100}{2} = 50$ A

(ii) $I_{avg} = \dfrac{100}{2} = 50$ A

$I_{rms} = \dfrac{100}{\sqrt{2}} = 70.7107$ A

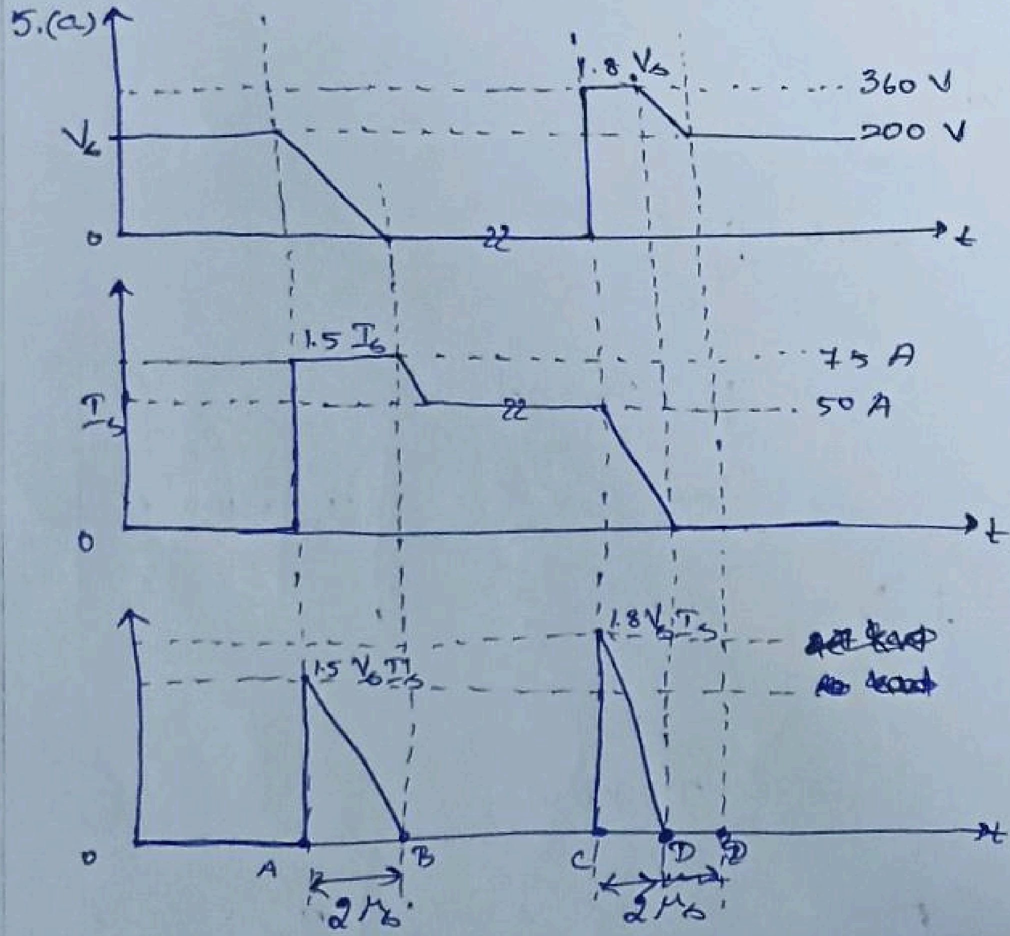
(b) $P = I_{rms}^2 r_d + I_{avg} V_{avg}$

$\Rightarrow 56.8 = 50^2 r_d + 31.831 V \rightarrow (1)$

$100 = 70.7107^2 r_d + 50 V \rightarrow (2)$

Solving (1) & (2), $\Rightarrow \boxed{r_d = 0.01\ \Omega\ ,\ V = 0.9956\ V}$

5.(a)



(b). ~~$P_{ON,pk} = 1.8\, V_s I_s = 24\ 1.5$~~

(b). $P_{ON,pk} = 1.5\, V_s I_s = 15\ kW$

$P_{OFF,pk} = 1.8\, V_s I_s = 18\ kW$

(c). $E_{ON} = \frac{1}{2} \times 2\, \mu s \times P_{ON,pk} = 15\ mJ$

$E_{OFF} = \frac{1}{2} \times 2\, \mu s \times P_{OFF,pk} = 18\ mJ$

$P_{sw,avg} = (E_{ON} + E_{OFF})\, f_{sw}$

$= (15+18) \times 10^{-3} \times 10 \times 10^{3} = 330\ W$

(d). $t_{AB} = 1\, \mu s = t_{CD}$

$\Rightarrow E_{ON} = \frac{1}{2} \times 1\, \mu s \times P_{ON,pk} = 7.5\ mJ$

$E_{OFF} = \frac{1}{2} \times 1\, \mu s \times P_{OFF,pk} = 9\ mJ$

$P_{sw,avg} = (E_{ON} + E_{OFF})\, f_{sw} = 165\ W$

$\Rightarrow$ Switching loss reduces as transition time reduces

6. $v(t) = \sum_{n=1}^{N} V_n \sin(n\omega t - \phi_n)$

$$V_{rms}^2 = \frac{1}{2\pi} \int_0^{2\pi} v^2(t)\, d(\omega t)$$

$$= \frac{1}{2\pi} \int_0^{2\pi} \left[\sum_{n=1}^{N} V_n \sin(n\omega t - \phi_n) \right]^2 d(\omega t)$$

$$\cancel{= \frac{1}{2\pi} \int_0^{2\pi} \left[\sum_{n=1}^{N} V_n^2 \sin^2(n\omega t - \phi_n) + \sum_{n\neq m} V_n V_m \sin(n\omega t - \phi_n) \right.}$$

$$V_{rms}^2 = \frac{1}{2\pi}\int_0^{2\pi}\Big[\sum_{n=1}^{N} V_n^2 \sin^2(n\omega t - \phi_n)$$
$$+ \sum_{n\neq m} V_n V_m \sin(n\omega t - \phi_n)\sin(m\omega t - \phi_m)\Big]\, d(\omega t)$$

$$V_{rms}^2 = \sum_{n=1}^{N} V_n^2 \cdot \frac{1}{2\pi}\int_0^{2\pi} \sin^2(n\omega t - \phi_n)\, d(\omega t)$$
$$+ \sum_{n\neq m} V_n V_m \cdot \frac{1}{2\pi}\int_0^{2\pi} \sin(n\omega t - \phi_n)\sin(m\omega t - \phi_m)\, d(\omega t) \rightarrow (1)$$

$$\frac{1}{2\pi}\int_0^{2\pi} \sin^2(n\omega t - \phi_n)\, d(\omega t) = \frac{1}{2} \rightarrow (2)$$

$$\frac{1}{2\pi}\int_0^{2\pi} \sin(n\omega t - \phi_n)\sin(m\omega t - \phi_m)\, d(\omega t) = 0 \rightarrow (3)$$

[$\sin(n\omega t - \phi_n)$ and $\sin(m\omega t - \phi_m)$ are orthogonal functions to each other]

(2) & (3) in (1):

$$V_{rms}^2 = \sum_{n=1}^{N} V_n^2\left(\frac{1}{2}\right) + \sum_{n\neq m} V_n V_m (0)$$

$$V_{rms}^2 = \sum_{n=1}^{N} V_n^2\left(\frac{1}{2}\right) = \sum_{n=1}^{N}\left(\frac{V_n}{\sqrt{2}}\right)^2 = \sum_{n=1}^{N} V_{n,rms}^2$$

$$\therefore \boxed{V_{rms}^2 = \sum_{n=1}^{N} V_{n,rms}^2}$$